

FUSION OF GLOBAL AND LOCAL TONE MAPPING FOR HIGH-DYNAMIC RANGE IMAGES

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Abstract

High dynamic range (HDR) image, when displayed on ordinary dynamic range devices, its visual effect will be weakened. To relieve this problem, this paper proposes a new method which combines global and local tone mapping together, to produce HDR image of better visual effect. The first step is to convert a HDR image to ordinary dynamic range using either global mapping or local mapping, depending on the mean intensity of the image. Then, Curvelet transform is applied to the mapped result to obtain a set of transform coefficients which are fused. Finally, taking advantages of both global and local mapping methods, a high-quality image is obtained. Experimental results show that the processed method provides images with more details and better visual effect, compared with other mapping methods such as Reinhard tonal mapping.

1 Introduction

High Dynamic Range (HDR) imaging can show the actual Dynamic Range of natural scene, which could generally meet nine orders of magnitude [1], which has a lot of details, realistic environment and clear feeling. But the current display device only has two orders of magnitude, which belongs to the low dynamic range device, and can't get to enrich the content of the HDR image fully displayed. If we want to display HDR images on monitor or printer, we need some mapping which can compress HDR image, and we call it Tone Mapping or tonal reconstruction. In essence, it is a tonal compression process, that is to say, it can remain the image details, high quality information (such as contrast and saturation) with the compression contrast ratio of HDR image. Literature [2] summarizes all kinds of high dynamic image tone mapping algorithm. Many scholars have made in-depth exploration for its research. In literature [3], tone mapping was introduced to computer graphics. In literature [4], Reinhard proposed tone mapping method which is based on photography method in 2002, however, it is easy to produce the halo phenomenon when two brightness point vary widely in the area using the local tone mapping method. In literature [5], Krik proposed a tone compression method based on visual perception against low-light scenarios, so we can calculate the middle of the perception of color vision and

scotopic scenario according to the spectral image data. In literature [6], a global mapping method was proposed which are consistent with the human visual model, and it can enhance the local contrast, at the same time it showed good properties in the aspect of human perception and psychology. In literature [7], HDR image visualization algorithm was proposed that based on image color appearance model (iCAM) framework, which can satisfy human visual perception of brightness and chromaticity in a wide range, but it will leads to the halo because of the spatial variability of iCAM, so the effect in some scenarios is poorer. In literature [8], Kuang proposed a new local algorithm based on hierarchical image color appearance model and bilateral filtering technology, color and detail reproduction are improved by this method. However, they considered the absolute change color of each pixel, without considering relative change between pixels, but human eyes are sensitive to the relative change. In addition, there also exists detail reproduction defect of bilateral filtering technology method. The above algorithm has a better effect for one part of HDR image which is the only global or local algorithm.

In order to get LDR images with high quality, this paper combines the advantage with disadvantages of the global algorithm and local algorithm. The first step is to map high dynamic range image in both global and local method by reinhard algorithm respectively. And then the Curvelet transform should be done to the result image. Because the global results are good in overall visual effect but it's hard to keep local contrast. While local results can keep good details, but this algorithm is complex and has a lot of parameters, it is hard to produce a better results image. So, we extract the global low frequency information as low frequency, and extract high frequency of the local and global information, then fusion them as high frequency, reconstruct image after combination.

The rest of this paper is organized as follows: section 2 describes the Global tone mapping. Section 3 presents the method of the Local tone mapping. Section 4 gives the Curvelet transform. The results and the evaluation are shown in section 5. Finally, section 6 concludes this paper.

2 Global tone mapping

In the global algorithm, the brightness of the image pixel logarithmic mean is the key to image scene. First of all, we calculate the value of whole image brightness of logarithmic mean. Formula for:

$$L_w^- = \frac{1}{N} \exp(\sum x, y \lg(\sigma + L_w(x, y))) \quad (1)$$

Among them, N is the total number of the image pixels, $L_w(x, y)$ is the location of (x, y) pixels. σ is a small positive number, which prevent the $L_w(x, y)$ is 0, and its logarithm is infinite. Average pixel brightness is always mapped to the key scene of a times, the equation can be written as:

$$L(x, y) = \frac{a}{L_w^-} L_w(x, y) \quad (2)$$

The above formula a is the adjustable parameter in the global tone mapping. Its value can be from 0 to 1. Experiments show that if image is normal brightness scenes, a is 0.18 usually, but if the image is high brightness scene, it needs to be reduced to below 0.18, the bright part in the image can be darker. Conversely, we need to increase the value of a , So we call it the proportionality constant of

brightness. Formula (3) will be mapped $L(x, y)$ to $L_d(x, y)$:

$$L_d = \frac{L(x, y)}{1 + L(x, y)} \quad (3)$$

The above formula can be extended as a controlled function:

$$L_d = \frac{L(x, y)(1 + \frac{L(x, y)}{L_{white}^2})}{1 + L(x, y)} \quad (4)$$

The default value of L_{white}^2 is L_{max} , while we can get good results. Then index transform to (4), Expressed as:

$$P_{ldr} = (\frac{hdr}{L_w(x, y)})^{satu} L_d \quad (5)$$

Among them, the $satu$ is exponential transform parameters. The function of exponential transform is extended high grayscale and compressed low grayscale.

3 Local tone mapping

Reinhard have proposed shading - exposure local tone mapping algorithm to compress the dynamic range [4], its basic idea is to find each pixel which does not exceed the threshold value around the contrast maximum surround field. It selects different a according to its brightness of around domain information. First, we need to find the target pixels radius of surrounding the domain. It meets the following formula:

$$|V(x, y, s_{max})| < \varepsilon \quad (6)$$

When s_{max} at get maximum value position (x, y) , corresponding to the maximum brightness around domain average pixel brightness is $V(x, y, s_{max})$. Local mapping algorithm formula:

$$L_d = \frac{L(x, y)}{1 + V(x, y, s_{max}(x, y))} \quad (7)$$

$$V(x, y, s_i) = \frac{|V_1(x, y, s_i) - V_1(x, y, s_{i+1})|}{2^\varphi a / s^2 + V_1(x, y, s_i)} \quad (8)$$

$$V_1(x, y, s_i) = L(x, y) \otimes G_i(h, v, \delta_i) \quad (9)$$

φ is sharpening parameters in the formula (8), it can enhance the edge of the image detail contrast. The experimental results show that the larger φ is, the more clearly on the detail in the image. Many experiments found $\varphi = 8$ is the best, if it is too big, the image will appear halo.

$G_i(h, v, \delta_i)$ is a gaussian filter, h, v are horizontal and vertical parameters, respectively, δ_i is Gaussian filter parameter. Finally, transformation of the dynamic range of compressed index, like equation (5), the image have displayed.

4 Curvelet transform

E. j. Candes and d. L. Donoho put forward curvelet transform theory [9], which is the first generation curvelet. The structure of the first generation curvelet thought curve by straight line of enough small block, then using local ridgelet to analyse its characteristics. In literature [10], the second generation curvelet was proposed. Fast curvelet transform algorithm is a simpler implementation that is easy to understand. In literature [11], E. j. Candes proposed two fast discrete methods based on the theory of the second generation of curvelet transform USFFT (Unequally-Space Fast Fourier Transform) algorithm and Wrapping (Wrapping of specially selected Fourier samples). The cartesian coordinates of two-dimensional discrete signal $f(n_1, n_2)$, $0 \leq n_1, n_2 < n$ as input, The definition of discrete curvelet transform:

$$C^D(j, l, k) = \frac{1}{L_{1,j} L_{2,j}} \sum_{n_1, n_2 \in p_j} \hat{f}[n_1 n_2 - n_1 \tan \theta_l] \hat{U}_j(n_1, n_2) \exp(i 2\pi (k_1 n_1 / L_{1,j} + k_2 n_2 / L_{2,j})) \quad (10)$$

Among them, $C^D(j, l, k)$ is discrete Curvelet transform, j, l, k are scale, direction and displacement parameters, respectively. $p_j = \{(n_1, n_2), 0 \leq n_1 < L_{1,j}, 0 \leq n_2 < L_{2,j}\}$

is support space of $\hat{U}_j(n_1, n_2)$, moreover $L_{1,j} \approx 2^j, L_{2,j} \approx 2^{j/2}$, $\hat{f}(n_1, n_2)$ is Fourier transform of discrete signal $f(n_1, n_2)$.

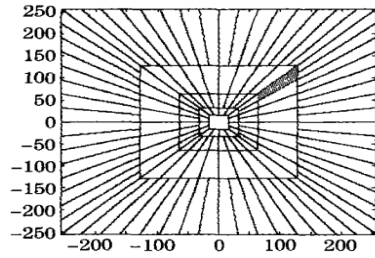


Figure 1 Curvelet frequency space area block diagram

Fusion rule: The low frequency part adopts the coarse scales of global image. High frequency part adopts the weighted result, when the piece of the value of the corresponding fine scale vary widely, maybe there has been noise, the weights using 0.5, 0.5, on the contrary, it is lost by global details, the weights using global 0.3, 0.7. So here it is necessary to make a threshold determination of the value. After a large number of experiments, it shows that the threshold set to 2000 is appropriate.

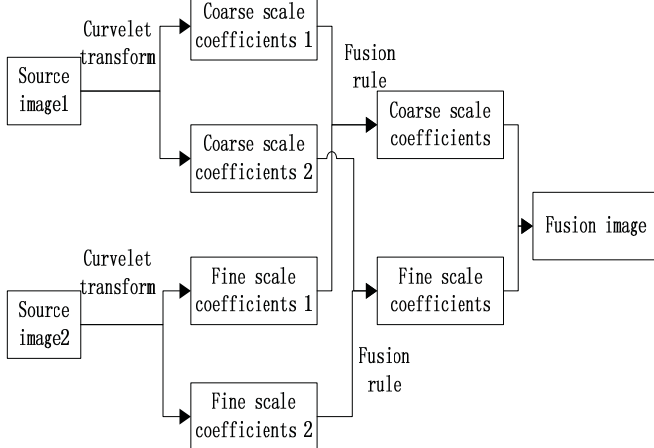


Figure 2 The fusion process

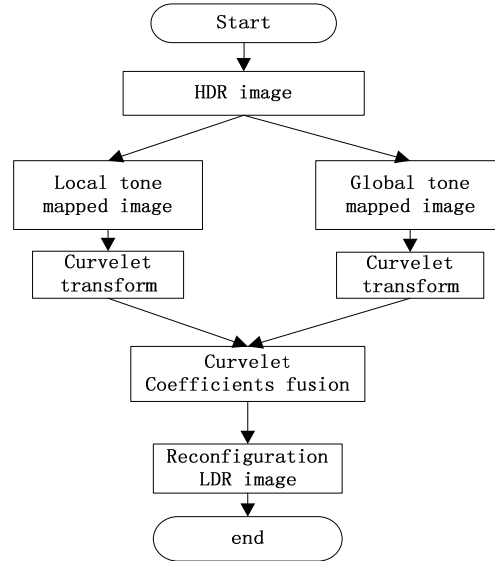


Figure 3 Algorithm process framework

5 The experimental results

In this paper, we have five groups of high dynamic range image (see figure 4), which were the significant differences in the content and dynamic range. The experimental results analysis includes subjective evaluation and objective evaluation.



Figure 4 Experimental sample figure

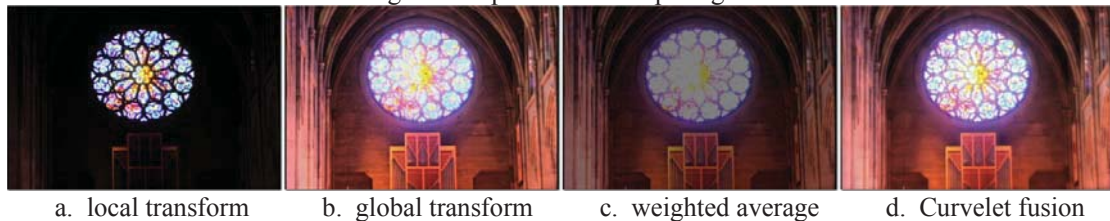


Figure 5 Rosette

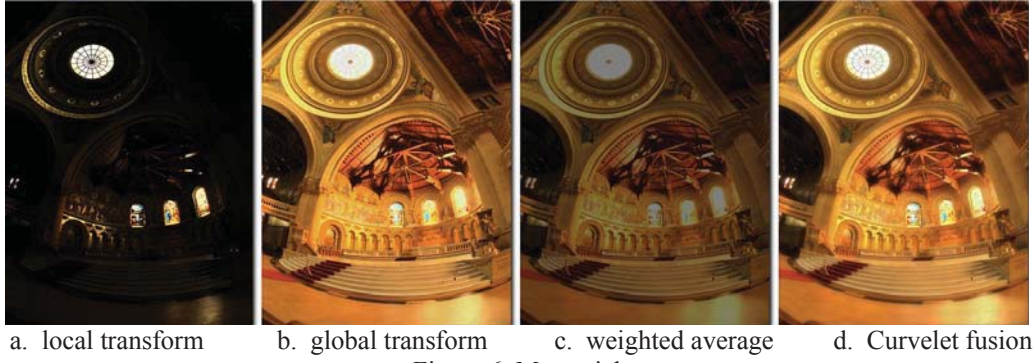


Figure 6 Memorial

5.1 The subjective evaluation

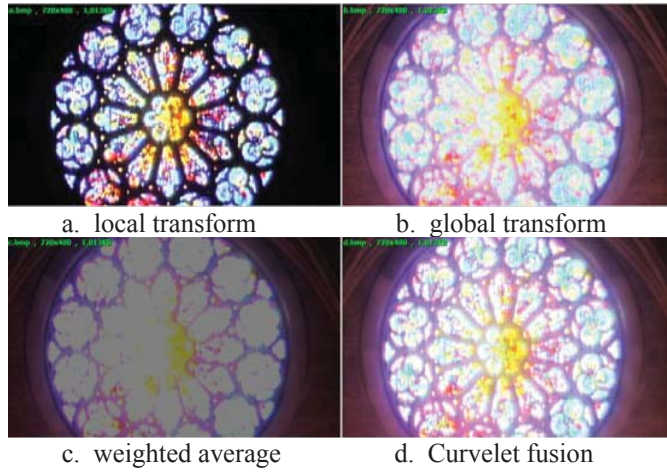


Figure 7 Rosette details analysis

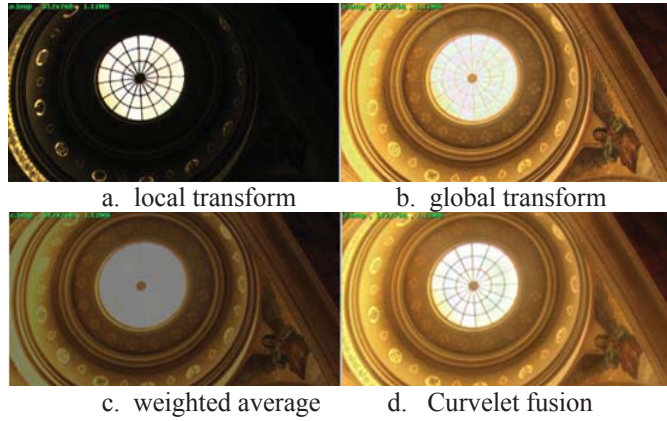


Figure 8 Memorial details analysis

It is easy to see from the experimental figure, the details of local transformation are very well, but because there are many values close or equal to zero in the dynamic range, which leads to dark images. The global transformation result has good visual perception, but the detail information is incomplete. The weighted average result is not enough clear obviously. Curvelet fusion is a good combination between local transformation and global transformation.

5.2 Objective evaluation

The experiment uses image information entropy, image resolution and peak signal to noise ratio as the image quality evaluation index. The larger the entropy of the image, the greater the amount of information in the image.

Information entropy formula is expressed as follows:

$$H = \sum_{i=0}^{255} -p_i \log(p_i) \quad (11)$$

The p_i is the probability of luminance equal to i in a image.

The larger the H is, the higher the image quality. Image resolution can describe the tiny details of the image. We can use the average gradient method to measure:

$$\Lambda = \frac{1}{(R-1)(C-1)} \sum_{i=0}^{R-1} \sum_{j=0}^{C-1} \sqrt{\frac{(x_{i,j} - x_{i+1,j})^2 + (x_{i,j} - x_{i,j+1})^2}{2}} \quad (12)$$

$x_{i,j}$ is the value of pixel of row i and column j in the image. R and C are the rows and columns of the image. The larger the Λ is, the higher the image quality.

Peak signal to noise ratio:

$$PSNR = 10 \times \log_{10} \left(\frac{MAX_I^2}{MSE} \right) = 20 \times \log_{10} \left(\frac{MAX_I}{\sqrt{MSE}} \right) \quad (13)$$

MSE is the mean square error, MAX_I is the biggest value of image tone. The larger the $PSNR$ is, the higher the image quality.

Table 1 objective evaluation of the five images

Table 1-1 The information entropy of the image

image	local	global	curvelet
Rosette	3.7373	5.9565	7.5784
Memorial	4.9379	7.7640	7.8059
Desk	6.3138	7.4959	7.4884
Atrium	6.5607	7.5666	7.4812
Window_series	4.2111	7.6172	7.6263

Table 1-2 The resolution of the image

image	local	global	curvelet
Rosette	4.0067	5.9893	4.6633
Memorial	2.7065	7.0396	5.3628
Desk	7.4963	5.7064	6.0090
Atrium	4.1003	6.0461	5.2329
Window_series	0.4369	1.8989	1.4472

Table 1-3 The PSNR of the image

image	local	global	curvelet
Rosette	28.6376	32.2817	34.4595
Memorial	30.5109	29.3408	31.6299
Desk	25.6842	29.1404	29.4946
Atrium	28.0478	27.5747	28.3621
Window_series	41.4244	39.2759	42.0204

From the Table 1, we know that curvelet fusion has the objective evaluation of the relative balance, make its subjective vision is very good. Table 1-1 shows that, curvelet fusion have the largest value in most of the time, for the rest of the time, they have a little bit smaller than the maximum value. Table 1-2 shows that, the value of the Λ of curvelet fusion are always between the maximum and minimum value. Table 1-3 shows that, the value of the PSNR of curvelet fusion are the largest.

6 Conclusion

In order to obtain a better effect, the Curvelet transform algorithm should be applied to the fusion reconstruction of mapping result, and then keep the coarse scale framework of the global algorithm part, the next work is to fuse the fine scale part of both global and local algorithm. Finally, the proposed method extract the partial algorithm information selectively. According to the experimental results, image which applied the new method keeps a higher detail information and has a better visual effect.

The algorithm can be applied in either forensic cases or photography, it can also be used in other image processing procedures, such as image feature extraction and enhancement processing. The future work is focus on enhancing the existing algorithms, and improving the image quality.

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